

Soft Spaces - Phase 3.2

Working hypothesis: P-Q coupling as the mechanism behind REAL-specific +/-E organization

Status: Hypothesis note preceding v31.0 mechanism probe

Baseline: Phase 3.1 frozen at Git tag phase-3.1-final

Purpose

Phase 3.1 established a reproducible organization of hotspot structure around paired positive and negative energies. The next task is no longer merely to detect or quantify that structure, but to identify a plausible operator-level mechanism that can distinguish REAL data from NULL controls.

This note defines one primary working hypothesis for Phase 3.2 and translates it into a small set of measurable quantities and falsifiable predictions.

Primary working hypothesis. The REAL-specific +/-E organization is not explained by spectral symmetry of H alone. It arises, at least in part, from a structured coupling between the complete degeneracy-closed P subspace and its complementary Q subspace.

The simplest operator diagnostic is therefore

$$\mathbf{P V Q}$$

and the natural second-order effective contribution is

$$\mathbf{P V Q (E - Q H Q)^{-1} Q V P}$$

1. P and Q as a dynamical partition

Let P denote the projector onto the complete degeneracy-closed subspace associated with the candidate hotspot group. Define $Q = I - P$. This partition separates the selected low-dimensional sector from all states outside it.

Within this decomposition, the perturbation V has three conceptually different roles:

Operator block	Interpretation
$P V P$	Perturbative action internal to the selected P sector
$P V Q$	Coupling from P into the complementary sector
$Q V P$	Return coupling from Q into P
$Q V Q$	Perturbative action internal to the complement

If REAL and NULL share approximately the same gross spectral organization of H, but differ systematically in P-to-Q coupling, then a purely spectral explanation becomes insufficient. The mechanism would instead be encoded in matrix elements that connect the selected subspace to nearby or resonantly relevant states outside it.

2. Why the second-order term is physically interesting

Eliminating Q perturbatively produces an energy-dependent effective operator acting inside P. At second order, the characteristic contribution has the form

$$\text{Sigma}_P(E) = P V Q (E - Q H Q)^{-1} Q V P$$

The resolvent $(E - Q H Q)^{-1}$ weights P-Q transitions according to their energy denominators. Consequently, two P sectors can have similar direct perturbative strength PVP but very different effective response if their coupling to Q states and their energy separations differ.

This gives a concrete route by which a hotspot can be enhanced: not simply because a state is spectrally special, but because its embedding in the surrounding Hilbert space produces an unusually coherent or strong second-order feedback into P.

3. Connection to the +/-E organization

Phase 3.1 indicates that candidate structures occur in strongly paired positive- and negative-energy sectors. The key Phase 3.2 question is therefore whether the effective P-Q coupling inherits the same pairing.

For a hotspot pair at +E and -E, define corresponding projectors P_+ and P_- . The mechanism hypothesis predicts that suitable scalar summaries of $P V Q$ or $\text{Sigma}_p(E)$ should themselves display stronger +/-E correspondence in REAL than in NULL.

Useful first diagnostics include:

1. Frobenius coupling strength: $\|P V Q\|_F^2$.
2. Spectral/operator norm of $P V Q$.
3. Trace- or norm-based strength of $\text{Sigma}_p(E)$.
4. Minimum and weighted energy gaps between P and the Q states that dominate the resolvent.
5. Correlation between the diagnostics evaluated for +E and -E partners, separately for REAL and NULL.

A useful first success criterion is not that every diagnostic differs. It is that at least one basis-invariant P-Q quantity reproduces the characteristic REAL > NULL +/-E organization robustly across the frozen Phase 3.1 candidates and seeds.

4. Falsifiable predictions

Prediction	Observation supporting it	Observation against it
H1	REAL has systematically stronger or more organized P-Q coupling than NULL	REAL and NULL are statistically indistinguishable
H2	+/-E partners show high correlation in a P-Q diagnostic in REAL	No +/-E correspondence beyond NULL
H3	Second-order effective strength explains more of hotspot structure than P-Q terms	Direct five-site P-Q terms add no information
H4	Controlled attenuation/scrambling of P-Q coupling weakens hotspot organization	Hotspot organization is unchanged after P-Q coupling is disrupted

5. Proposed v31.0 mechanism probe

The first Phase 3.2 program should be deliberately diagnostic rather than exploratory. It should reuse the frozen Phase 3.1 Hamiltonian construction, candidate definition, REAL/NULL generation, degeneracy handling and seed conventions wherever possible.

Suggested filename: v31_0_mechanism_probe.py

Primary question: Can a basis-invariant P-to-Q coupling metric explain the REAL-specific +/-E organization already observed in Phase 3.1?

For each frozen candidate group and its +/-E partner, v31.0 should compute P and Q and record a compact diagnostic vector containing PVP strength, PVQ strength, relevant P-Q energy gaps, a regularized second-order effective term, and the corresponding +/-E pair statistics. The same pipeline should then be applied unchanged to REAL and NULL.

Numerical care is required near poles of the resolvent. The first implementation should therefore use an explicitly stated regularization or pseudoinverse convention, and sensitivity to that choice should be checked before interpreting $\text{Sigma}_p(E)$ quantitatively.

6. Decision logic for Phase 3.2

A positive result would be evidence that the hotspot phenomenon is controlled by how the P sector is embedded in and coupled to the remainder of Hilbert space, rather than by the local spectrum alone. That would motivate v31.1 as a causal intervention: alter or scramble only the P-Q coupling structure and test whether the hotspot signal follows.

A negative result would also be informative. If P-Q diagnostics fail to distinguish REAL from NULL, Phase 3.2 should not be rescued by indiscriminately adding more statistics. The mechanism search should move to another operator-level candidate while preserving the Phase 3.1 baseline.

Working statement

The central Phase 3.2 hypothesis is that the observed REAL-specific +/-E organization is a property of subspace embedding and perturbative inter-sector coupling. The operator $P V Q$ is the minimal diagnostic of that embedding, while $P V Q (E - Q H Q)^{-1} Q V P$ provides the natural second-order effective mechanism to test.

This document is a hypothesis and experiment-design note, not a claim of an established physical mechanism. Its purpose is to define what Phase 3.2 must attempt to confirm or falsify.